

The NS Sphere Block from Free Fermion plus Super-Virasoro and from Two Virasoro Algebras

Machine note

20 August 2026

Abstract

We derive the two-Virasoro expansion of the sphere four-point block of four bottom NS superconformal primaries, using exactly the slot order and odd three-form sign of `Human Notes/SCblock.tex`. The auxiliary fermion is put in the identity state at every external puncture. With the ungraded pairing on $F_{1/2} \otimes SCA_{1/2}$, its block is one and the enlarged block equals the super-Virasoro block. Resolving the same block in the $\text{Vir} \oplus \text{Vir}$ branching basis gives a sum over $n \in \frac{1}{2}\mathbb{Z}$. The human-note endpoint convention supplies the essential factor $(-1)^{2n}$. A direct PBW computation agrees with the resulting formula through even level two and odd level $3/2$. The nonchiral crossing equation is unchanged because this sign is common to the whole odd chiral block and therefore squares to one.

1 Conventions and the ordered NS block

We use the ordinary central charge and momentum convention of the human note,

$$Q = b + b^{-1}, \quad c = \frac{3}{2} + 3Q^2, \quad h(P) = \frac{1}{2} \left(\frac{Q^2}{4} - P^2 \right). \quad (1)$$

The Liouville continuum is reached by $P = ip$; it is useful to keep P generic while proving the algebraic identity. All external states ϕ_i are even bottom components. The ordered correlator is

$$\langle \phi_4(\infty) \phi_3(1) \phi_2(z) \phi_1(0) \rangle. \quad (2)$$

Let $\xi_A = \mathbf{L}_{-A} \phi_h$ be an ordered NS PBW state, let $|A|$ be its level, and let $p(A)$ be its fermion parity. The inverse super-Gram matrix is $\mathbf{G}_h^{AB}$. We strip the common leading power and define the two parity blocks by

$$\begin{aligned} F_a^{4321}(h; z) &= z^{h-h_1-h_2} \mathcal{F}_a^{4321}(h; z), \\ \mathcal{F}_a^{4321}(h; z) &= \sum_{\substack{|A|=|B| \\ p(A)=p(B)=a}} z^{|A|} \rho_a(\phi_4, \phi_3, \xi_A) \mathbf{G}_h^{AB} \rho_a(\xi_B, \phi_2, \phi_1), \quad a \in \{0, 1\}. \end{aligned} \quad (3)$$

Thus $\mathcal{F}_0$ has integer powers and $\mathcal{F}_1$ has half-odd-integer powers.

The sign which matters below is fixed by the component table in the human note. For even primaries,

$$\rho_1(G_{-1/2} \phi_1, \phi_2, \phi_3) = +1, \quad \rho_1(\phi_1, G_{-1/2} \phi_2, \phi_3) = +1, \quad \boxed{\rho_1(\phi_1, \phi_2, G_{-1/2} \phi_3) = -1}. \quad (4)$$

The last minus sign is retained throughout this note. Some production sphere-block code uses +1 in the third entry; that is a different chiral convention and differs from the present odd block by an overall minus sign.

2 The ungraded tensor-product pairing

Adjoin an auxiliary NS Majorana fermion with

$$\{\psi_r, \psi_s\} = \delta_{r+s,0}, \quad \psi_r^\dagger = -\psi_{-r}, \quad r \in \mathbb{Z} + \frac{1}{2}. \quad (5)$$

The internal enlarged module is

$$\mathcal{M}_{\text{NS}}(h) = \mathcal{V}_{\text{F}_{1/2}} \otimes \mathcal{V}_{\text{SCA}_{1/2}}(h). \quad (6)$$

The essential choice is the prime, ungraded pairing

$$\boxed{\langle\langle u \otimes x \mid v \otimes y \rangle\rangle' = \langle\langle u \mid v \rangle\rangle_{\text{F}} \langle\langle x \mid y \rangle\rangle_{\text{SCA}}.} \quad (7)$$

There is no Koszul sign crossing the two tensor factors. This differs from the standard graded tensor-product pairing.

For product states the normalized three-form is the one printed in the current human note,

$$\widehat{\rho}_a(u_1 \otimes x_1, u_2 \otimes x_2, u_3 \otimes x_3) = (-1)^{|x_1|(|u_2|+|u_3|)+|x_2||u_3|} \rho^{\text{F}}(u_1, u_2, u_3) \rho_a(x_1, x_2, x_3). \quad (8)$$

At the external punctures choose $\widehat{\phi}_i = \mathbf{1}_{\text{F}} \otimes \phi_i$. In the product PBW basis the Gram matrix and its inverse factor as ordinary tensor products because of Eq. (7). Equation (8) has no crossing sign at either sphere trinion: the external SCA states and the external fermion vacua are even. Moreover,

$$\rho^{\text{F}}(\mathbf{1}, \mathbf{1}, u) = \rho^{\text{F}}(u, \mathbf{1}, \mathbf{1}) = 0 \quad \text{unless} \quad u = \mathbf{1}. \quad (9)$$

It follows coefficient by coefficient that

$$\widehat{\mathcal{F}}_a^{4321}(h; z) = \mathcal{F}_{\text{F}}^{1111}(z) \mathcal{F}_a^{4321}(h; z), \quad \boxed{\mathcal{F}_{\text{F}}^{1111}(z) = 1, \quad \widehat{\mathcal{F}}_a^{4321} = \mathcal{F}_a^{4321}.} \quad (10)$$

Unlike the genus-two theta graph, there is no multivariable polarized convolution: the free-fermion identity block kills every nonvacuum auxiliary state.

3 The same enlarged block in the double-Virasoro basis

For $b \neq 1$, the human-note generators give $\text{Vir}^{(1)} \oplus \text{Vir}^{(2)} \subset \text{F} \oplus \text{SCA}$. Their central charges are

$$c^{(i)} = 1 + 6(Q^{(i)})^2, \quad (b^{(1)})^2 = \frac{2b^2}{1-b^2}, \quad (b^{(2)})^{-2} = \frac{2}{b^2-1}, \quad Q^{(i)} = b^{(i)} + (b^{(i)})^{-1}. \quad (11)$$

In the notation of the human note, the NS branching rule is

$$\mathcal{M}_{\text{NS}}(h(P)) \cong \bigoplus_{n \in \frac{1}{2}\mathbb{Z}} \mathcal{V}_{c^{(1)}}(h_n^{(1)}(P)) \otimes \mathcal{V}_{c^{(2)}}(h_n^{(2)}(P)), \quad h_n^{(1)} + h_n^{(2)} = h(P) + 2n^2. \quad (12)$$

In a square-root-free parametrization useful for computation,

$$h_n^{(1)}(P) = \frac{(Q^{(1)})^2}{4} - \frac{(P + 2nb)^2}{2 - 2b^2}, \quad h_n^{(2)}(P) = \frac{(Q^{(2)})^2}{4} - \frac{(P + 2n/b)^2}{2 - 2b^{-2}}. \quad (13)$$

The branch primary $v_n(P)$ has parity $2n \pmod{2}$ and ungraded norm $N_n(P) = \langle\langle v_n \mid v_n \rangle\rangle'$.

Now the reason for Eq. (7) becomes explicit. If I, J and I', J' are ordinary Virasoro partitions, then

$$\begin{aligned} \langle\langle L_{-I}^{(1)} L_{-J}^{(2)} v_n \mid L_{-I'}^{(1)} L_{-J'}^{(2)} v_n \rangle\rangle' \\ = N_n(P) (G_{h_n^{(1)}}^{(1)})_{II'} (G_{h_n^{(2)}}^{(2)})_{JJ'}. \end{aligned} \quad (14)$$

The inverse therefore factorizes with no parity operator left on the tube. The graded product pairing would insert a state-dependent crossing sign and would not give Eq. (14) in this basis.

Let $\mathcal{V}(c; d_4, d_3, d_2, d_1; d; z) = 1 + O(z)$ denote the reduced ordinary Virasoro sphere block in the same slot order. Define the two unnormalized branching vertices

$$T_{43}(n) = \widehat{\rho}_{2n \bmod 2}(v_{4,0}, v_{3,0}, v_n), \quad T_{21}(n) = \widehat{\rho}_{2n \bmod 2}(v_n, v_{2,0}, v_{1,0}). \quad (15)$$

We now use exactly the branching-coefficient notation of the human note:

$$B_a(h_1, h_2, h_3; n_1, n_2, n_3) = \frac{\widehat{\rho}_a(v_{1,n_1}, v_{2,n_2}, v_{3,n_3})}{\|v_{1,n_1}\| \|v_{2,n_2}\| \|v_{3,n_3}\|}. \quad (16)$$

The external branch vectors have unit norm. Hence the two coefficients at the sphere endpoints are

$$B_a(h_4, h_3, h; 0, 0, n) = \frac{T_{43}(n)}{\|v_n\|}, \quad B_a(h, h_2, h_1; n, 0, 0) = \frac{T_{21}(n)}{\|v_n\|}. \quad (17)$$

Ordinary Virasoro Ward identities act independently in the two factors. Inserting the inverse of Eq. (14) into the four-point block gives

$$\boxed{\mathcal{F}_a^{4321}(h(P); z) = \sum_{\substack{n \in \frac{1}{2}\mathbb{Z} \\ (-1)^{2n} = (-1)^a}} z^{2n^2} B_a(h_4, h_3, h; 0, 0, n) B_a(h, h_2, h_1; n, 0, 0) \mathcal{V}_n^{(1)}(z) \mathcal{V}_n^{(2)}(z)}. \quad (18)$$

where

$$\mathcal{V}_n^{(i)}(z) = \mathcal{V}\left(c^{(i)}; h_0^{(i)}(P_4), h_0^{(i)}(P_3), h_0^{(i)}(P_2), h_0^{(i)}(P_1); h_n^{(i)}(P); z\right). \quad (19)$$

The product of branching coefficients is independent of a rephasing of v_n . If the same square-root branch for $\|v_n\|$ is used at both ends, it can equivalently be written

$$B_a(h_4, h_3, h; 0, 0, n) B_a(h, h_2, h_1; n, 0, 0) = \frac{T_{43}(n) T_{21}(n)}{N_n(P)}. \quad (20)$$

Thus $T_{43} T_{21} / N_n$ is only a phase-independent representation of the human-note product $B_a B_a$, not a replacement notation for the branching coefficient.

4 The human-note endpoint sign and the explicit blow-up formula

We now translate the branching coefficients in Eq. (18) to the factorized product l used in the human note and in the blow-up literature. Only in the following product formula, $m, k', k \in \mathbb{Z}$ are integer Fermi-sea labels; the internal sphere label is $k = 2n$. For $\sigma, \tau = \pm 1$, set

$$x_{\sigma\tau} = \alpha + \sigma P' + \tau P, \quad r_{\sigma\tau} = \frac{m + \sigma k' + \tau k}{2}. \quad (21)$$

For $r \geq 0$, define

$$\begin{aligned} s_e(x, r) &= 2^{-r^2/2} \prod_{\substack{i, j \geq 1, \ i+j \leq 2r \\ i+j \equiv 0 \pmod{2}}} [x + (i-1)b + (j-1)b^{-1}], \\ s_o(x, r) &= 2^{-r(r+1)/2} \prod_{\substack{i, j \geq 1, \ i+j \leq 2r+1 \\ i+j \equiv 1 \pmod{2}}} [x + (i-1)b + (j-1)b^{-1}], \end{aligned} \quad (22)$$

and continue to negative r by

$$s_e(x, r) = (-1)^r s_e(Q - x, -r), \quad s_o(x, r) = s_o(Q - x, -r). \quad (23)$$

With $\text{Int}(x) = \text{sgn}(x) \lfloor |x| \rfloor$,

$$l(\alpha, m \mid P', k', P, k) = \begin{cases} \prod_{\sigma, \tau = \pm 1} s_e(x_{\sigma\tau}, r_{\sigma\tau}), & m + k + k' \text{ even}, \\ \prod_{\sigma, \tau = \pm 1} s_o(x_{\sigma\tau}, \text{Int } r_{\sigma\tau}), & m + k + k' \text{ odd}. \end{cases} \quad (24)$$

For the branch label $n \in \frac{1}{2}\mathbb{Z}$, the norm is the identity specialization with the integer Fermi-sea label $2n$:

$$N_n(P) = s_e(2P, 2n) s_e(-2P, -2n). \quad (25)$$

For the sphere endpoints define

$$\begin{aligned} L_{43}(n) &= l\left(\frac{Q}{2} + P_3, 0 \mid P_4, 0, P, 2n\right), \\ L_{21}(n) &= l\left(\frac{Q}{2} + P_2, 0 \mid P, 2n, P_1, 0\right). \end{aligned} \quad (26)$$

The human-note odd table (4) fixes the ordered endpoint translation:

$$\boxed{T_{43}(n) = (-1)^{2n} L_{43}(n), \quad T_{21}(n) = L_{21}(n).} \quad (27)$$

This is not an optional phase of v_n : a rephasing of v_n occurs at both endpoints and cancels against N_n . It is the relative sign between placing the odd fixed-parity tensor in slot three and in slot one. Direct branching-vector calculations verify Eq. (27) for $n = 0, \pm \frac{1}{2}$, and for the one-active-leg cases $n = 1, \frac{3}{2}$; the general factor is fixed by the same parity-homogeneous three-form normalization.

Equations (17) and (27) give the explicit product of the two human-note branching coefficients:

$$\boxed{B_a(h_4, h_3, h; 0, 0, n) B_a(h, h_2, h_1; n, 0, 0) = (-1)^{2n} \frac{L_{43}(n) L_{21}(n)}{N_n(P)}.} \quad (28)$$

Substituting Eq. (28) in the boxed block relation (18) reproduces the square-root-free expression used in the numerical check. Since $(-1)^{2n} = (-1)^a$ within a fixed block, the sign is a common minus on the odd chiral block and is one on the even block. Equation (18) is first derived for generic $b \neq 1$ and generic momenta; the $b = 1$ result is its analytic limit after the two Virasoro factors have been combined.

4.1 Analytic level-1/2 sign check

The first branch primaries are

$$v_{\pm\frac{1}{2}}(P) = \left[G_{-1/2} + \left(\frac{Q}{2} \pm P \right) \psi_{-1/2} \right] \phi, \quad (29)$$

and their ungraded norms are

$$N_{+\frac{1}{2}}(P) = -P(Q + 2P), \quad N_{-\frac{1}{2}}(P) = P(Q - 2P). \quad (30)$$

The external fermion identities kill the $\psi_{-1/2}$ term. Hence Eq. (4) gives $T_{43}(\pm\frac{1}{2}) = -1$, $T_{21}(\pm\frac{1}{2}) = +1$, while $L_{43}(\pm\frac{1}{2}) = L_{21}(\pm\frac{1}{2}) = 1$. The double-Virasoro answer at the first odd level is therefore

$$[z^{1/2}] \mathcal{F}_1 = - \left(\frac{1}{N_{+\frac{1}{2}}} + \frac{1}{N_{-\frac{1}{2}}} \right) = - \frac{4}{Q^2 - 4P^2} = \boxed{-\frac{1}{2h(P)}}. \quad (31)$$

Direct super-Virasoro sewing gives the same result: $(-1)(+1)/\|G_{-1/2}\phi\|^2 = -1/(2h)$. Dropping $(-1)^{2n}$ gives $+1/(2h)$, the opposite convention.

5 Direct low-order PBW check

The reproducible calculation is

`Code/check_ns_sphere_double_virasoro.py.`

It constructs the finite- c NS PBW basis and Gram matrix directly, evaluates both descendant three-forms from Ward identities, and compares with Eq. (18). Ordinary Virasoro Gram matrices are evaluated independently through level two. The generic test point is

$$\begin{aligned} b &= 1.37, & Q &= 2.09992700729927018, \\ c &= 14.7290803079546091, & P &= 0.47, \\ (P_1, P_2, P_3, P_4) &= (0.23, -0.31, 0.41, -0.19), & h(P) &= 0.440761679498108683. \end{aligned} \quad (32)$$

The reduced coefficients are

$2L$	direct human sign	double Virasoro	absolute error	without $(-1)^{2n}$
0	1.0000000000000000	1.0000000000000000	0	1.0000000000000000
1	-1.134399888323653	-1.134399888323653	5.04×10^{-17}	1.134399888323653
2	0.178198040229849	0.178198040229849	1.54×10^{-17}	0.178198040229849
3	-0.484955260912690	-0.484955260912690	1.75×10^{-16}	0.484955260912690
4	0.120205847599768	0.120205847599768	7.79×10^{-17}	0.120205847599768

Thus the check reaches even level two, including the new $n = \pm 1$ branch primaries, and odd level $3/2$. The maximum discrepancy is 1.75×10^{-16} . Omitting the human endpoint sign flips every odd coefficient and fails at the first possible order.

6 Crossing equation

For four bottom NS Liouville primaries the nonchiral direct-channel decomposition in the project convention is

$$G_{4321}^{(s)}(z, \bar{z}) = \int_0^\infty \frac{dp}{\pi} [C_{12p} C_{34p} F_0^{4321}(p; z) F_0^{4321}(p; \bar{z}) + \tilde{C}_{12p} \tilde{C}_{34p} F_1^{4321}(p; z) F_1^{4321}(p; \bar{z})]. \quad (33)$$

The crossed channel is obtained by $(P_1, P_2, P_3, P_4; z) \mapsto (P_3, P_2, P_1, P_4; 1 - z)$.

Equation (18) changes the odd chiral convention by $F_1^{\text{human}} = -F_1^{\text{production}}$, while leaving F_0 unchanged. The same sign occurs in the antiholomorphic block, so

$$F_1^{\text{human}}(z) F_1^{\text{human}}(\bar{z}) = F_1^{\text{production}}(z) F_1^{\text{production}}(\bar{z}). \quad (34)$$

Consequently crossing is exactly insensitive to this simultaneous chiral sign flip. It does, however, test the relative coefficient of the even and odd *nonchiral* families. We therefore performed a new computation in which the two continuum channels were evaluated independently, without averaging, rescaling, or imposing equality. The driver is

`Code/stress_ns_crossing_extended.py.`

It works at the self-dual point $b = 1$, where the physical structure constants used here are implemented. Changing only the block central charge would not define a check at another b , so no such artificial b -scan was made.

The blocks use the direct functional c -recursion at $c = 27/2$, with exact global $\mathfrak{osp}(1|2)$ leaves and no local- z or elliptic- q truncation. We used 80-digit block arithmetic, 35-digit structure constants, $p \leq 5$, and 32-point Gauss–Legendre quadrature. For each of the three momentum quadruples

$$\left(\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{3}{5}\right), \quad (0.20, 0.45, 0.70, 0.35), \quad (0.80, 0.15, 0.55, 0.40), \quad (35)$$

we evaluated seven real points $z = 0.2, 0.3, \dots, 0.8$ and three complex points $z = 0.3 + 0.2i, 0.5 + 0.25i, 0.7 + 0.2i$. Thus the main ledger contains 90 raw channel comparisons. With

$$R = \frac{|G^{(s)}(z, \bar{z}) - G^{(t)}(1 - z, 1 - \bar{z})|}{\max(|G^{(s)}(z, \bar{z})|, |G^{(t)}(1 - z, 1 - \bar{z})|)}, \quad (36)$$

the maximum over the ten points is

momenta in Eq. (35)	$N = 8$	$N = 10$	$N = 12$	wrong $G_0 - G_1, N = 12$
first	6.80×10^{-5}	9.43×10^{-6}	1.32×10^{-6}	6.32×10^{-2}
second	7.25×10^{-5}	1.01×10^{-5}	1.42×10^{-6}	6.55×10^{-2}
third	7.92×10^{-5}	1.12×10^{-5}	1.58×10^{-6}	7.34×10^{-2}

Here N is the maximum accumulated physical null level in the recursion. The worst order-twelve point is the third case at $z = 0.8$. At $z = 0.5$ the three residuals are respectively 7.11×10^{-11} , 2.61×10^{-11} , and 2.02×10^{-10} ; the largest residual among the complex points is 2.36×10^{-8} . The maximum channel drift from $N = 10$ to $N = 12$ is 9.57×10^{-6} , showing that the edge residual is dominated by recursion truncation. At real z , the largest spurious imaginary part is 7.46×10^{-37} .

As a negative control, the even and odd momentum integrals were also retained separately. At $N = 12$, the largest even-only and odd-only crossing residuals are 3.65×10^{-2} and 8.83×10^{-1} , whereas their physical sum gives the table above. Replacing that sum by the deliberately wrong nonchiral combination $G_0 - G_1$ produces the last column. This is a test of the physical even/odd assembly, not a test of the chiral phase in Eq. (34).

Finally, for the first momentum set and $z = 0.2, 0.5, 0.8, 0.5 + 0.25i$, we varied the continuum integration at fixed $N = 12$:

integration	maximum R	maximum channel change from previous row
$p \leq 4$, 24 points	1.319268×10^{-6}	—
$p \leq 5$, 32 points	1.319270×10^{-6}	3.25×10^{-9}
$p \leq 6$, 40 points	1.319261×10^{-6}	6.01×10^{-11}

At the last resolution, $R(0.5) = 1.40 \times 10^{-12}$ and $R(0.5 + 0.25i) = 5.79 \times 10^{-12}$. The integration drift is therefore far below the order-twelve edge residual. These are convergence diagnostics, not certified error bounds for the infinite recursion.

The raw ledgers are

```
DataSet/ns_liouville_crossing_extended_fresh.json,
DataSet/ns_liouville_crossing_extended_p4_q24.json,
DataSet/ns_liouville_crossing_extended_p6_q40.json.
```

They support nonchiral crossing and its physical even/odd coefficients, but the direct PBW comparison in the previous section remains what fixes the human-note chiral sign $(-1)^{2n}$.

7 Conclusion

With the ungraded pairing, the auxiliary-fermion/SCA Gram matrix and the double-Virasoro Gram matrix both factor without a parity insertion. The auxiliary identity block is one, so the two computations of the enlarged sphere block give Eq. (18). The only nontrivial ordered-sphere sign is $(-1)^{2n}$, inherited from the human note's third-slot odd tensor. It passes the direct low-order computation; deleting it produces the opposite odd block. The extended crossing scan strongly checks the physical nonchiral even/odd assembly and remains satisfied under recursion and integration refinement, but it cannot distinguish the two overall chiral odd-block sign conventions.